ML-Based Condition Monitoring and Remaining Life Prediction of Turbofan Engines

ML Lab Mini-Project
Phase 1: Current Implementation and Future Work

Madaka Shreyaj Naidu – 240962594
Advait Balachandar – 240962600
Dipen Batharaju – 240962574

V Semester, Section A | Batch 2028
B.Tech. in CSE (AI & ML)
School of Computer Engineering
Manipal Institute of Technology
Manipal Academy of Higher Education
Academic Year 2026

1. Project Overview

The project, titled “ML-Based Condition Monitoring and Remaining Life Prediction of Turbofan Engines”, aims to develop a machine-learning-based predictive maintenance framework using the NASA C-MAPSS (Commercial Modular Aero-Propulsion System Simulation) dataset. The dataset contains multivariate measurements collected over successive operating cycles of turbofan engines. The overall goal is to use these measurements to estimate Remaining Useful Life (RUL) and support earlier maintenance decisions.

The project synopsis defines the final system as a comparison of Random Forest, XGBoost, LightGBM and Long Short-Term Memory (LSTM) models for RUL prediction. The proposed workflow includes data preprocessing, feature engineering, RUL target construction, normalization, sliding-window sequence generation, model training, evaluation using MAE and RMSE, and comparative visualization.

At the current stage, the team has completed the initial data preparation pipeline and an initial Random Forest implementation. The work completed so far establishes the foundation required for the remaining model-development and evaluation stages.

2. Project Objectives

- Develop machine-learning models capable of estimating the Remaining Useful Life of turbofan engines.
- Preprocess and analyze sensor data to identify useful degradation-related attributes.
- Compare ensemble machine-learning models, including Random Forest, XGBoost and LightGBM.
- Develop an LSTM model capable of learning temporal degradation patterns from engine-cycle sequences.
- Evaluate the models using suitable regression metrics and comparative visualizations.
- Use the final predictions to provide useful engine-health and maintenance information.

3. Dataset and Data Preparation

The project uses the NASA C-MAPSS turbofan engine degradation dataset. The original dataset is provided in space-separated TXT files. Each record contains an engine identifier, operating cycle, three operating settings and 21 sensor measurements. The project uses the FD001 dataset for the current prepared-model implementation, while the conversion code is structured to process FD001, FD002, FD003 and FD004.

The first implementation stage converts the raw TXT files into structured CSV files with explicit column names. The conversion process also constructs RUL values for training records and derives a maintenance-oriented binary label using an RUL threshold of 30 cycles. This creates a consistent machine-readable format for downstream modelling.

3.1 Work Completed by Dipen Batharaju

Dipen completed the raw dataset conversion stage. The original NASA C-MAPSS TXT files were read using whitespace separation, empty columns were removed, and the records were assigned descriptive column names. The conversion pipeline was prepared for all four C-MAPSS subsets: FD001, FD002, FD003 and FD004.

For the training data, RUL was derived by subtracting the current operating cycle from the maximum cycle reached by the corresponding engine. A binary maintenance label was also generated using the condition RUL $\leq$ 30 cycles. The processed records were then exported as CSV files for subsequent modelling.

3.2 Work Completed by Madaka Shreyaj Naidu

Shreyaj completed the preprocessing and feature-selection stage used for the initial model. The raw converted dataset was examined and the attributes considered useful for engine-health prediction were identified. The prepared dataset reduces the original sensor space to a smaller set of relevant variables, making the data more suitable for model development.

The selected attributes include the operating cycle and sensors representing temperature, pressure, rotational speed and engine performance/efficiency. The current project documentation identifies Sensor 2 (T24), Sensor 3 (T30), Sensor 4 (T50), Sensor 7 (P30), Sensor 8 (Nf), Sensor 9 (Nc), Sensor 15 (BPR), Sensor 17 (htBleed), Sensor 20 (W31) and Sensor 21 (W32) among the retained sensor measurements.

3.3 Current Data Pipeline

Stage	Input	Output
Raw data	NASA C-MAPSS TXT files	Raw engine-cycle records
Conversion	Space-separated TXT data	Structured CSV files with named columns
RUL construction	Engine cycles	RUL values for training records
Preprocessing	Converted CSV	Cleaned/selected modelling features
Current modelling dataset	Prepared FD001 data	Feature matrix and RUL/maintenance target

4. Initial Machine Learning Implementation

Advait completed the initial Random Forest stage using the prepared dataset supplied after Shreyaj's preprocessing. The current implementation treats engine health as a binary classification problem by converting RUL into two maintenance categories: Safe when more than 30 cycles remain and Maintenance Needed when 30 or fewer cycles remain.

The Random Forest implementation uses an 80:20 training-testing split and trains a Random Forest Classifier with 100 estimators. The current implementation also evaluates the classifier using accuracy, precision, recall and F1-score, and generates a confusion matrix and feature-importance visualization.

4.1 Work Completed by Advait Balachandar

Advait implemented and executed the first machine-learning model using the prepared FD001 dataset. This provides the first working baseline for the project and demonstrates that the selected sensor attributes contain useful information for distinguishing engines approaching failure from engines with more remaining useful life.

4.2 Initial Result

Metric / Class	Current result
Overall accuracy	96.05%
Safe – Precision	0.97
Safe – Recall	0.98
Maintenance Needed – Precision	0.88
Maintenance Needed – Recall	0.84

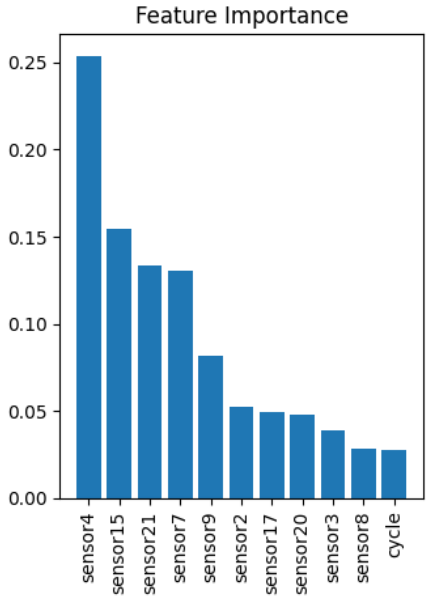


Figure 1. Random Forest feature-importance plot for the selected engine sensor attributes.

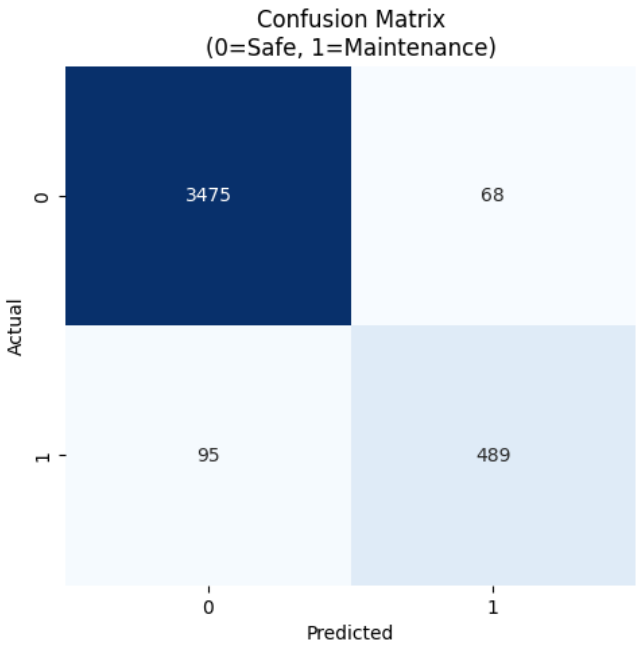


Figure 2. Random Forest evaluation: confusion matrix for the Safe and Maintenance Needed classes.

The initial result is encouraging. The notebook also generated a confusion matrix and a feature-importance plot, reproduced above, to inspect classification behaviour and the relative contribution of the selected engine attributes.

4.3 Important Scope Clarification

There is a distinction between the current implementation and the final proposed methodology. The current Random Forest stage uses a binary maintenance label derived from RUL. The final project is intended to return an RUL estimate and compare Random Forest, XGBoost, LightGBM and LSTM using regression-oriented evaluation. The binary maintenance interpretation can subsequently be derived from the predicted RUL.

5. Work Remaining and Equal Distribution

The remaining implementation has not yet been started. To distribute the work equally across the three members, the following responsibilities are assigned. Each member will handle a major technical component while contributing to the final integration and documentation.

5.1 Dipen Batharaju – Gradient-Boosting Models and Model Tuning

- Implement XGBoost for RUL regression using the prepared feature set.
- Implement LightGBM for RUL regression.
- Experiment with important hyperparameters such as tree depth, number of estimators, learning rate and sampling parameters.
- Generate predictions for the test data and calculate MAE and RMSE.
- Create comparison plots/tables for Random Forest, XGBoost and LightGBM.
- Provide the tuned tree-based models for integration into the final comparison.

5.2 Madaka Shreyaj Naidu – Temporal Modelling with LSTM

- Prepare the selected sensor data for sequential/time-series modelling.
- Create sliding-window sequences from consecutive engine cycles.
- Define the appropriate RUL target for each sequence.
- Normalize/reshape the input data for LSTM training.
- Design, train and validate the LSTM model.
- Tune sequence length, hidden units, dropout and training parameters as required.
- Evaluate the LSTM using MAE and RMSE and provide actual-vs-predicted plots.

5.3 Advait Balachandar – Evaluation, Explainability and Final Integration

- Extend the current Random Forest implementation from the initial classification baseline toward the final RUL-regression comparison where appropriate.
- Build a common evaluation framework for all final models.
- Compare models using MAE and RMSE, with additional supporting metrics/plots where useful.
- Perform feature-importance analysis and investigate SHAP-based explanations for the final model.
- Translate predicted RUL into practical engine-health/maintenance information.
- Integrate the selected model outputs, visualizations and maintenance interpretation into the final project workflow.
- Support the final dashboard/interface and presentation.

6. Proposed Final Methodology

The completed system will follow a sequential pipeline from raw engine data to an interpretable maintenance output. The methodology follows the structure defined in the project synopsis while incorporating the work already completed.

Step	Planned activity	Expected output
1	Load and validate C-MAPSS data	Consistent engine-cycle dataset
2	Preprocess and select useful sensors	Reduced and suitable feature set
3	Construct RUL target	Continuous RUL target for regression
4	Feature engineering and scaling	Model-ready numerical features
5	Random Forest regression	Baseline RUL predictions
6	XGBoost and LightGBM	Gradient-boosted RUL predictions
7	Sliding-window generation	Temporal input sequences
8	LSTM training	Sequence-based RUL predictions
9	Hyperparameter tuning	Improved model configurations
10	Model evaluation	MAE/RMSE comparison
11	Explainability	Important sensor/degradation factors
12	Maintenance interpretation	Actionable engine-health output
13	Final integration	Complete predictive-maintenance system

7. Evaluation Strategy

The final models will be evaluated primarily as RUL regression models. Mean Absolute Error (MAE) will measure the average magnitude of prediction error, while Root Mean Squared Error (RMSE) will give greater weight to large prediction errors. Actual-versus-predicted plots will be used to visually assess how closely the predictions follow the true RUL values.

The final comparison will consider both predictive accuracy and practical usefulness. A model with low numerical error but unstable behaviour on late-life engine cycles may not be preferable to a slightly less accurate model that provides more consistent maintenance warnings. The final selection will therefore consider the quantitative results together with feature importance, interpretability and the intended maintenance application.

8. Expected Final Outcome

At the end of the remaining implementation, the project is expected to provide a complete machine-learning pipeline for turbofan engine condition monitoring and RUL prediction. The system will take engine operating-cycle and sensor measurements as input, estimate the remaining useful life, compare the performance of multiple machine-learning approaches, and convert the prediction into useful maintenance information.

The final system will contain both conventional ensemble models and a temporal deep-learning model. This will allow the team to study whether the additional temporal information captured by an LSTM provides an advantage over tree-based models. The comparison will also establish which model is most suitable for the available C-MAPSS data and the project's predictive-maintenance objective.

9. Phase-wise Plan of Execution

Phase	Activities	Responsible members
Phase 1 – Completed	TXT-to-CSV conversion; preprocessing; feature selection; initial Random Forest classification.	Dipen, Shreyaj, Advait
Phase 2	XGBoost, LightGBM and Random Forest regression/tuning.	Dipen + team integration
Phase 3	Sliding-window preparation and LSTM modelling.	Shreyaj + team integration
Phase 4	Common evaluation, model comparison, explainability and maintenance interpretation.	Advait + team
Phase 5	Final integration, visualizations, report and presentation.	All three members

10. Current Status Summary

Component	Status	Work completed / planned output
Dataset conversion	Completed	NASA C-MAPSS TXT files converted into structured CSV files; conversion pipeline supports FD001–FD004.
Dataset preprocessing	Completed	Prepared modelling dataset created; relevant sensor attributes identified and retained.
RUL preparation	Completed / used in current pipeline	RUL information is available and used to derive the maintenance-oriented target.
Random Forest	Initial implementation completed	Random Forest classifier trained on the prepared FD001 dataset.
XGBoost	Planned	Implement and evaluate an XGBoost RUL prediction model.
LightGBM	Planned	Implement and evaluate a LightGBM model.
LSTM	Planned	Create sliding-window sequences and train an LSTM for temporal RUL prediction.
Hyperparameter tuning	Planned	Tune tree and sequence models and select suitable configurations.
Final evaluation	Planned	Compare models using appropriate RUL metrics, mainly MAE and RMSE, with supporting visualizations.
Explainability / maintenance interpretation	Planned	Analyze important features and translate predicted RUL into useful maintenance information.
Final integration and presentation	Planned	Combine the best model, visualizations and maintenance output into the final project system.

11. Conclusion

The project has successfully completed its initial implementation stage. The team has converted the raw NASA C-MAPSS data into usable CSV files, prepared a reduced modelling dataset through preprocessing and sensor selection, and implemented an initial Random Forest classifier. The current classifier achieves 96.05% accuracy on the reported test split, with particularly strong performance for the Safe class.

However, the current result is only an initial milestone. The final project requires the team to move from the current maintenance-oriented classification baseline toward the RUL regression framework described in the synopsis. The next stages will therefore focus on XGBoost, LightGBM, LSTM-based temporal modelling, hyperparameter tuning,

quantitative comparison, explainability and final system integration. These steps will allow the project to progress from a working baseline to a complete predictive-maintenance solution.

12. Reference Project Documentation

The project scope, objectives and proposed methodology follow the submitted mini-project synopsis: “ML-Based Condition Monitoring and Remaining Life Prediction of Turbofan Engines.” The synopsis specifies NASA C-MAPSS as the primary dataset and proposes Random Forest, XGBoost, LightGBM and LSTM models, with preprocessing, feature engineering, sliding-window processing and MAE/RMSE-based evaluation.